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Deep Attention SMOTE: Data augmentation with a learnable interpolation factor for imbalanced anomaly detection of gas turbines

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Highlights

- A new hybrid re-sampling scheme is proposed to overcome the class imbalance issue.
- A feature space with inter-class separability is learned for data augmentation.
- An attention module is designed to learn adaptive interpolation factors.
- The adaptive factors generate high-quality synthetic samples in the feature space.
- The model is verified using a real-world gas turbine dataset from an Asian airline.

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Deep Attention SMOTE: Data augmentation with a learnable interpolation factor for imbalanced anomaly detection of gas turbines

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Declaration of competing interest

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Abstract

Anomaly detection of gas turbines faces the significant challenges of data imbalance and inter-class overlap. In this paper, we develop a novel data augmentation method, namely deep attention synthetic minority over-sampling technique with the Encoder-Decoder (DA-SMOTE-ED), which serves as a key step in anomaly detection. To reduce the risk of generating noise data, on one hand, the DA-SMOTE-ED leverages an Encoder-Decoder to learn a class-separable feature space to weaken the effect of inter-class overlap. On the other hand, an attention module is applied to assign proper interpolation factors to generate synthetic samples that stay off the aggregation area of normal samples. Moreover, synthetic samples are generated in the learnable feature space, mapped back to the original space, and merged with under-sampled samples to form the balanced data set. Finally, compared to previous re-sampling methods, experimental results validate that the developed method achieves superior performance for anomaly detection.

Keywords:
gas turbine, imbalanced data, inter-class overlap, data augmentation, anomaly detection

1. Introduction

Gas turbines are the core component of aircraft, and have a significant influence on flight safety. To prevent some minor anomalies from evolving into catastrophic faults in the flight, one of the most effective ways is to detect these anomalies before takeoff. Early detection provides airlines with enough time to exclude these anomalies which can not only economize on unscheduled downtime but also ensure flight safety[1]. Therefore, it is increasingly important to develop accurate anomaly detection methods for gas turbines[2].

The anomaly detection methods for gas turbines are mainly divided into three categories, including model-based, traditional machine learning-based, and deep learning-based methods. On one hand, model-based methods have good interpretations for the anomaly detection process, but require much domain-specific knowledge. Especially for complex equipment, it is extremely difficult to construct accurate physical models to match actual gas turbines and reflect the dynamic response of the damage[3][4]. On the other hand, traditional machine learning-based methods are easy to implement, yet these shallow models require handcrafted features due to their limited ability to process raw data[5][6]. If handcrafted features cannot characterize gas turbines' running status, detection results will be seriously affected.

In contrast, deep learning-based methods can automatically extract useful and informative features from raw data[7][8], which omits handcrafted feature design and complex modeling procedures. Thus, they have tremendous potential in the field of anomaly detection[9][10]. However, the direct application of deep learning still faces the two significant challenges: data imbalance and inter-class overlapping[11], as depicted in Fig.1(a). The reasons are that on one hand, gas turbines fly in normal status at most time, and then generally slowly evolve into abnormal status. Most of the collected data is normal, while only a fraction of that is abnormal. That is, the collected data is highly imbalanced. On the other hand, abnormal data is similar to the normal data right before anomaly, resulting in inter-class overlapping.

Up to now, re-sampling methods are still common ways to solve the aforementioned challenges, including under-sampling, data augmentation, and hybrid methods[12]. Suppose that the data set consists of N normal samples and A abnormal samples. To balance the data set, traditional under-sampling methods generally remove $N-A$ normal samples (Fig.1(b)), while data augmentation methods usually generate $N-A$ synthetic abnormal samples (Fig.1(c)). Hybrid methods adopt a combination of the above two methods to ensure that samples from different classes have the same number (denoted as K). More specifically, hybrid methods simultaneously remove $N-K$ normal samples and generate $K-A$ abnormal samples. Obviously, according to $N-K<N-A$, hybrid methods can preserve

more information compared with only under-sampling methods. According to $K - A < N - A$, hybrid methods can reduce the risk of introducing noise samples compared with only data augmentation methods. Therefore, hybrid methods are selected to balance the data set in this paper.

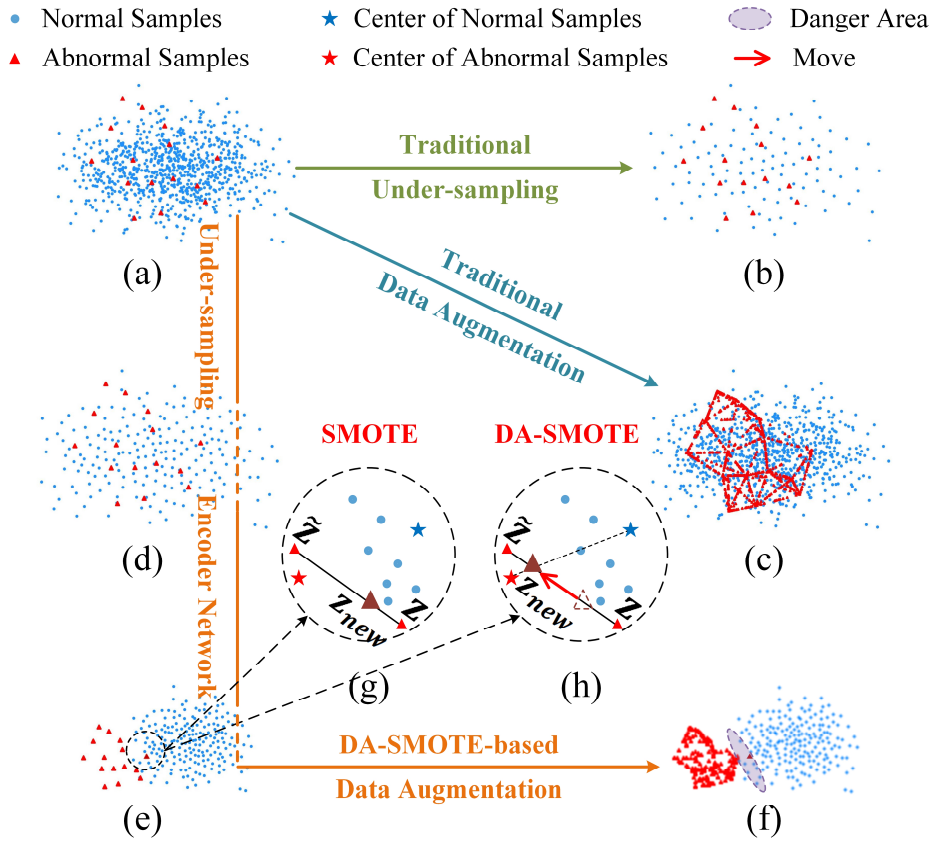

Fig. 1 2D sketch maps of (a) all samples, (b) under-sampled samples, (c) data-augmented samples, when using traditional re-sampling methods; the effect of our hybrid-sampling method with 3 key steps, including (d) under-sampling, (e) feature projection, and (f) DA-SMOTE based data augmentation; and zoom-in comparison of synthetic sample generation using (g) SMOTE and (h) DA-SMOTE, where z is an abnormal sample, $\tilde{z}$ is a randomly selected abnormal neighbor of z, and z_{new} is a synthetic sample.

More notably, hybrid methods generally adopt the synthetic minority over-sampling technique (SMOTE) as the data augmentation method[13][14]. The SMOTE generates synthetic samples by randomly interpolating along the lines between abnormal samples and their nearest abnormal neighbors[15]. More specifically, suppose that z is an abnormal sample, and $\tilde{z}$ is one of its k nearest neighbors. First, an interpolation factor α is randomly chosen from a uniform distribution $U(0, 1)$, and then the synthetic abnormal sample z_{new} is calculated according to $z_{new} = z + \alpha(\tilde{z} - z)$. As depicted in Fig.1(g), the selection of α is random, and z_{new} may fall into the danger area and become the noise sample, where the danger area represents the area in which the number of abnormal samples is far less than that of normal samples[16].

To reduce the risk of generating noise data, this paper develops a novel data augmentation method, known as DA-SMOTE-ED, which serves as a key step in balancing the data set. The DA-SMOTE in the DA-SMOTE-ED employs an attention module to learn an appropriate interpolation factor according to data distribution, and the learnable factor keeps the synthetic sample as far from the danger area as possible, as depicted in Fig.1(f).

Moreover, the full steps are described as follows. First, the raw data set is grouped into K clusters, and cluster centers are combined with abnormal data to construct the under-sampled data set, as depicted in Fig.1(d). Second, the under-sampled data set is mapped into a learnable feature space with better inter-class separability, as shown in Fig.1(e). Third, the DA-SMOTE generates synthetic samples in the feature space. As depicted in Fig.1(h), synthetic samples continually move towards the center of abnormal samples while move away from that of normal samples in the training process, thereby reducing the generation of noise data. Fourth, synthetic samples are mapped back to the original space, and then put together with under-sampled data set, to serve as the balanced data set.

The main contributions can be summarized as follows.

1) A new attention-based hybrid re-sampling scheme is proposed to overcome the class imbalance issue, in which the clustering-based under-sampling method can alleviate the degree of class imbalance and retain more normal information compared to traditional under-sampling methods.

2) A feature space with better inter-class separability is learned via a self-attention-based Encoder-Decoder, which reduces the overlap of samples belonging to different classes and provides favorable conditions for data augmentation. In addition, the triplet-center loss is used together with the reconstruction loss to increase intra-class

compactness.

3) To further alleviate the effect of imbalanced data, a novel data augmentation method is developed to generate high-quality minority samples in the learnable feature space. Significantly, an attention module is designed to learn an adaptive interpolation factor from the data distribution. In the training phase, the interpolation factor makes the synthetic minority sample dynamically move away from the aggregation area of the majority samples, thereby reducing the risk of generating noise data.

4) An intelligent anomaly detection framework is proposed to infer the health status of gas turbines, in which multiple self-attention modules can jointly extract different aspects of temporal features from multi-dimensional time series. Furthermore, the effectiveness of the framework is evaluated using a real-world gas turbine data set from an Asian airline.

The rest of this paper is organized as follows. Section 2 reviews the related work about popular deep learning-based anomaly detection methods and re-sampling methods. Section 3 elaborates on our hybrid re-sampling method and applies it to the anomaly detection of gas turbines. Section 4 constructs comparative experiments and discusses experimental results. Finally, Section 5 summarizes the main conclusions.

2. Related Work

2.1. Deep learning for gas turbine anomaly detection

In recent years, the deep learning-based anomaly detection method for gas turbines has received unprecedented attention. For example, Yan et al.[9] utilizes a stacked denoising auto-encode to extract high-level features from raw monitoring data, and then combines the extreme learning machine to detect anomalies. Khani et al.[17] applies filters to preprocess the raw data set, and then combines deep convolutional neural network (CNN) to detect the health status of gas turbines. Bai et al.[18] adopts a group of long short-term memory (LSTM) networks to learn complex temporal features from monitoring data detected by multi-sensors. Nor et al.[19] combines a deep Bayesian neural network (DBNN) with the Shapley additive interpretation to build an explainable anomaly detection framework, and the effectiveness of the framework is verified on real-world data sets. The aforementioned works require a balanced training set. However, this is not the case in practice.

2.2. Traditional re-sampling methods for imbalanced classification

To obtain a balanced training set, the first method (under-sampling) is to reduce the number of majority samples, which faces the issue of valuable information loss[20]. For example, Lin et al.[21] applies a clustering technique to preprocess the raw data and selects the nearest neighbors of cluster centers to represent the majority samples. Koziarski[22] uses a criterion, namely mutual class potential, to guide the process of under-sampling. The second method (data augmentation) is to increase the number of minority samples, which faces the risk of generating noise samples[23]. For example, Han et al.[24] calculates borderline samples belonging to the minority class, and then only over-samples these samples. Douzas et al.[25] uses K-Means to divide the raw data into different clusters, and then generates minority samples in each cluster, respectively. The third method (hybrid re-sampling) is a combination of the under-sampling method and the data augmentation method. For example, Batista et al.[26] applies the SMOTE to generate minority samples, and then removes the noise samples and boundary samples. Sowah et al.[27] combines the cluster under-sampling with the Sigma nearest over-sampling based on the convex combination, yielding pretty good accuracy and reliability in the imbalanced classification.

3. Proposed Anomaly Detection Framework

This section elaborates on the research motivations, the proposed anomaly detection framework, and the developed DA-SMOTE-ED in our hybrid re-sampling method.

3.1. Research motivations

Gas turbine anomaly detection is an extremely important way to ensure customer safety and save the airlines operating costs. However, deep learning-based methods face at least the following three challenges.

Data imbalance: For gas turbines, the abnormal data is very limited and far less than normal data. In the training stage, the classification loss is dominated by normal data. That is, if normal data is classified correctly, the classification loss will be very small. In this case, the classifier pays more attention to normal data, and tends to classify all data as normal data[28]. However, this is not allowed in engineering.

Inter-class overlap: The abnormal data is similar to normal data (right before anomaly) and overlaps with it. In this case, the synthetic abnormal data may be closer to normal data but still has the label “abnormal”, which will mislead the training of deep learning models[29]. Therefore, a significant task is to reduce the degree of inter-class overlap before data augmentation.

Complex temporal pattern: The monitoring data of gas turbines belong to the typical multi-dimensional time series. The dependencies, trends, and correlations among different performance parameters are complex and unclear[30]. That is, it is a challenging problem to extract complex temporal patterns from monitoring data.

3.2. Overview of the proposed anomaly detection framework

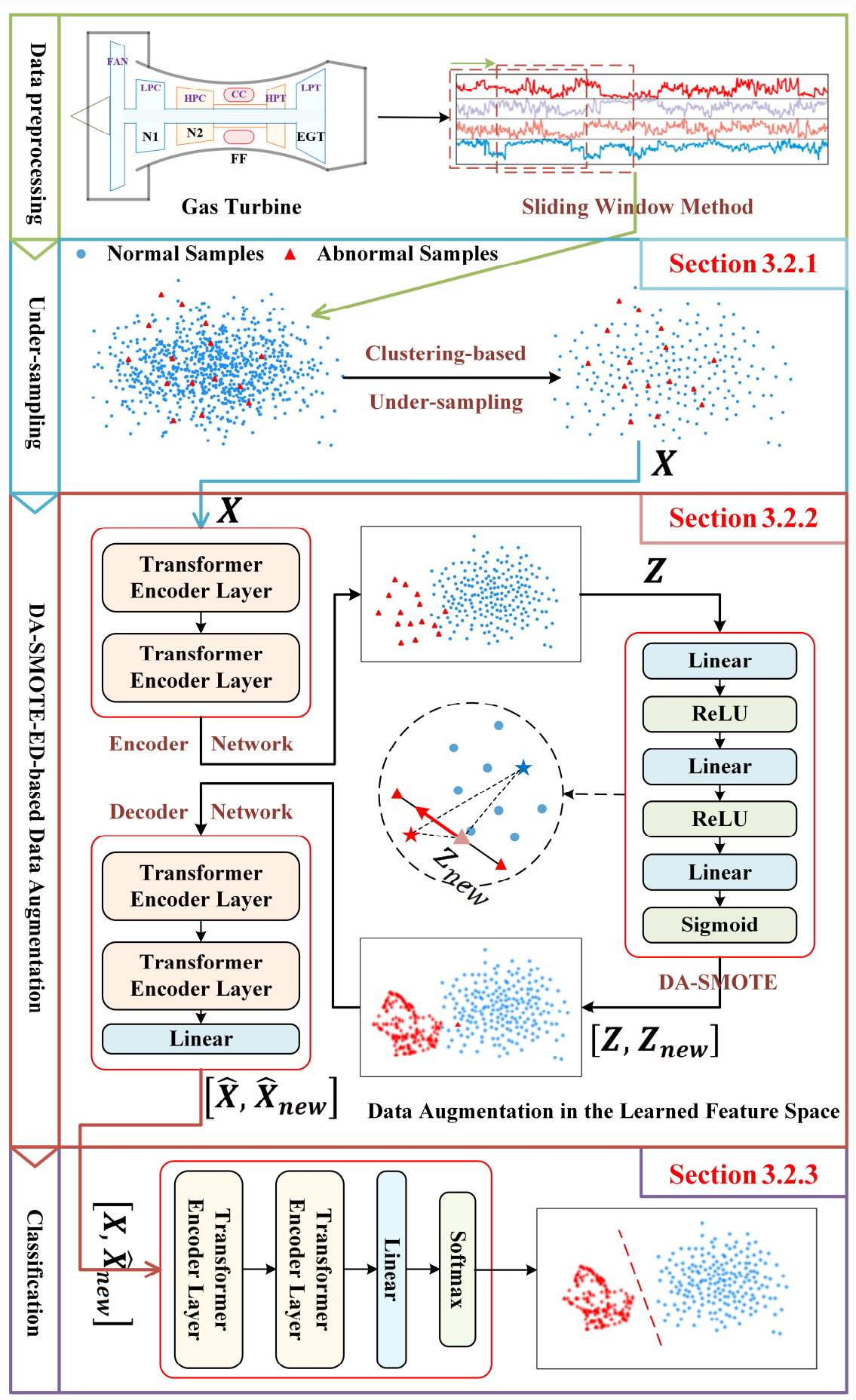

Fig. 2 The overview of the proposed anomaly detection framework.

Aiming at class imbalance and inter-class overlap of the multi-dimensional time series, this paper proposes a novel attention-based hybrid re-sampling scheme and applies it to constitute the anomaly detection framework with a self-attention classification network. Moreover, the overall process of the proposed anomaly detection framework is depicted in Fig.2.

Without loss of generality, the set of normal samples in the training set is denoted as $X_n = \{x_1, x_2, \cdots, x_N\}$, where N is the number of normal samples. Meanwhile, the set of abnormal samples is denoted as $X_a = \{x_1, x_2, \cdots, x_A\}$, where A is the number of abnormal samples.

3.2.1. Clustering-based under-sampling

The clustering-based under-sampling uses a proper clustering technique to divide normal samples into different clusters, and then selects the cluster centers to represent normal data, which can reduce the degree of class imbalance to a certain extent. The K-Means is selected as the clustering technique due to its effectiveness and simple implementation[20][21]. More specifically, X_n is divided into K clusters obtaining groups of similar samples[31], and the set of cluster centers is $C = \{c_1, c_2 \cdots, c_K\}$, where c_j is the j-th cluster center. Then, C is put together with X_a to construct the under-sampled training set X, that is, $X = C \bigcup X_a$. It is noticeable that the quantitative difference of normal samples with abnormal samples falls from $N - A$ to $K - A$.

3.2.2. DA-SMOTE-ED-based data augmentation

As depicted in Fig.2, the DA-SMOTE-ED consists of three basic substructures: an Encoder network (for mapping the under-sampled data into a feature space), an attention module (for learning an adaptive interpolation factor), and a Decoder network (for mapping the synthetic samples back to the original space). More specifically, first, the triplet-center loss and reconstruction loss are jointly used to train the DA-SMOTE-ED, and more details are given in **Section 3.3**. Second, the trained Encoder network maps X into the feature space with inter-class separability ($Z = Encoder(X)$). Third, the trained attention module selects a proper interpolation factor to generate high-quality synthetic data Z_{new}. Fourth, the trained Decoder network maps Z_{new} back to the original space ($\hat{X}_{new} = Decoder(Z_{new})$). Finally, the synthetic samples $\hat{X}_{new}$ is merged with the under-sampled training set X to form the balanced training set $[X, \hat{X}_{new}]$.

3.2.3. Anomaly classification

After obtaining the balanced training set, the self-attention network (SANet) is selected as the classifier, because it is more suitable to deal with the multi-dimensional time series compared with traditional recurrent neural networks[32][33]. More specifically, first, $[X, \hat{X}_{new}]$ is fed into the SANet, and its Softmax output is $O \bigcup O_{new}$, that is, $O \bigcup O_{new} = SANet([X, \hat{X}_{new}])$. Second, the binary cross-entropy loss is calculated according to the following formula $\mathcal{L}_C = \frac{1}{|O \bigcup O_{new}|} \sum_{o_i \in O \bigcup O_{new}} [y_i log(o_i) + (1 - y_i) log(1 - o_i)]$, where o_i and y_i are the Softmax output and label of the i-th sample respectively. Third, the SANet is trained using the binary cross-entropy loss. Finally, the trained SANet can be used to directly infer whether test samples are abnormal.

3.3. Developed DA-SMOTE-ED for Intelligently generating synthetic abnormal samples

This subsection introduces the basics of the classical SANet, and then focuses on the developed DA-SMOTE-ED-based data augmentation in our hybrid-sampling method.

3.3.1. Architecture of the Classical SANet

In recent years, the SANet is proven to have extraordinary potential in the multi-dimensional time series analysis[32][34]. As depicted in Fig.3(a), the SANet consists of multiple Transformer Encoder layers (i.e., a multi-head self-attention mechanism, two short-cuts & layer normalizations, and a fully connected feed-forward layer) and a linear layer with the Softmax. Furthermore, Fig.3(b) shows that a multi-head self-attention mechanism consists of multiple parallel self-attention modules[35], in which the calculation process of the scaled dot-product attention is depicted in Fig.3(c).

Taking the i-th self-attention module for example, first, X is mapped to the i-th subspace by three linear projections, and the outputs are three self-attention matrices Q_i, K_i, V_i, which are expressed by

$$Q_i = XW_i^Q, \quad K_i = XW_i^K, \quad V_i = XW_i^V \tag{1}$$

where W_i^Q, W_i^K, W_i^V are parameter matrices of three linear projections. Second, the self-attention weight matrix A_i is calculated according to the following formula

$$A_i = softmax(Q_i {K_i}^T / \sqrt{d_i}) \tag{2}$$

where d_i is the dimension of Q_i. Significantly, A_i can enhance important features and weaken useless features, thus making the inference result more accurate[36]. Third, the obtained information of the i-th self-attention module is denoted as H_i, which is expressed by

$$H_i = A_i V_i \tag{3}$$

Finally, all features that are extracted by different self-attention modules are concatenated and projected to obtain the final feature H, which is expressed by

$$H = Concat(H_1, H_2, \cdots, H_h) W \tag{4}$$

where h is the number of self-attention modules, and W is the parameter matrix of the linear projection.

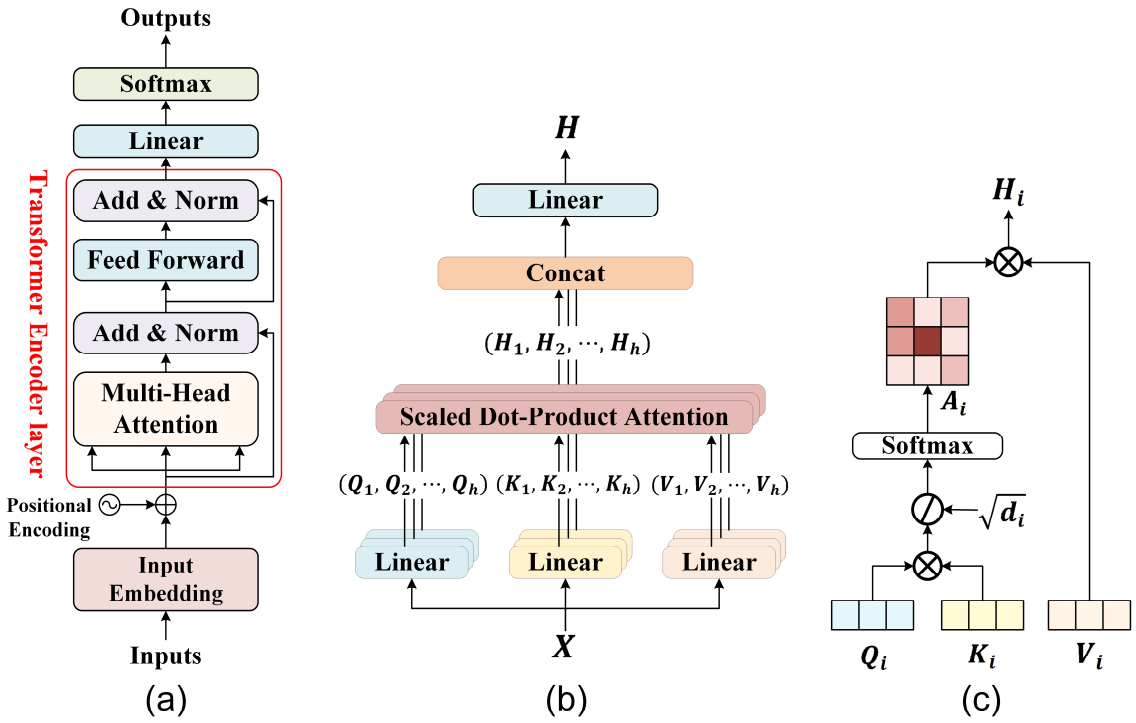

Fig. 3 (a) The SANet architecture, (b) multi-head self-attention mechanism, (c) scaled dot-product attention.

3.3.2. Architecture and key procedure of the DA-SMOTE-ED

In the original space, abnormal samples highly overlap with normal samples, and direct data augmentation may lead synthetic samples to fall into the danger area. To alleviate the issue, the following two aspects can be considered: (1) mapping original samples into a separable feature space so as to shrink the danger area; (2) learning an adaptive interpolation factor from data distribution so as to move the synthetic samples away from the danger area. Based on the above analysis, this paper develops a novel data augmentation method (DA-SMOTE-ED) that integrates an Encoder network, a DA-SMOTE, and a Decoder network, as depicted in Fig.4(a). Moreover, the developed DA-SMOTE-ED is trained in an end-to-end manner.

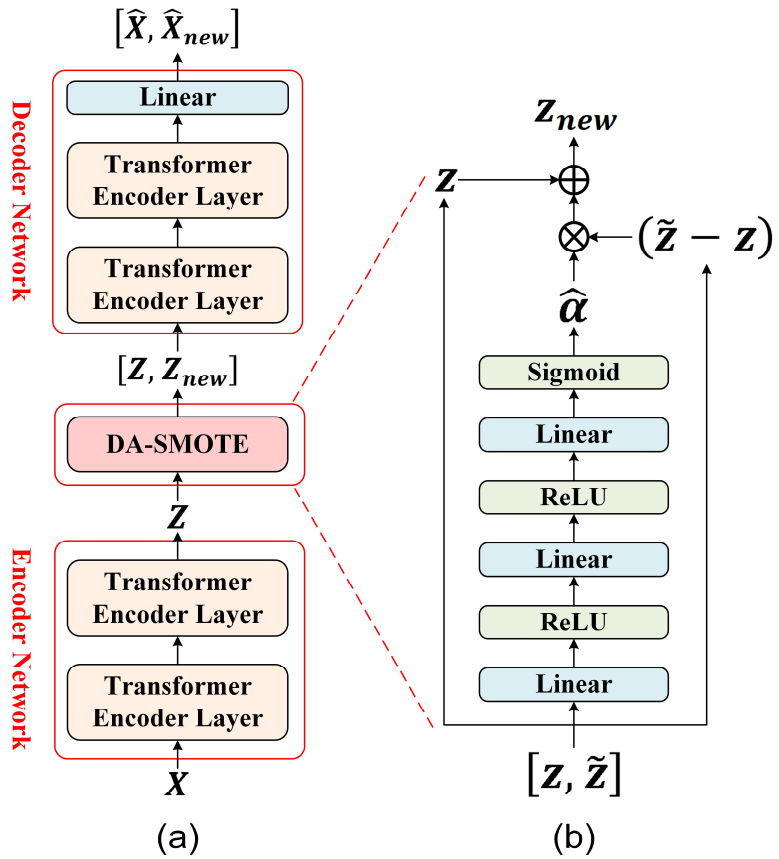

Fig. 4 (a) The developed DA-SMOTE-ED architecture, (b) the building unit entitled DA-SMOTE.

Encoder Network: To shrink the danger area, the developed method adopts an Encoder-Decoder to learn a separable feature space, in which samples from different classes are far away from one another. In this case, the synthetic samples are also far away from normal samples, thereby reducing the risk of falling into the danger area.

The Encoder network aims to map the original data into the separable feature space, and is composed of multiple Transformer Encoder layers, as depicted in Fig.4(a). The reason is that the multi-head self-attention mechanism in the Transformer Encoder layer can map the multi-dimensional time series into different subspaces, assign different weights to select important temporal features, and then merge them to form more complex and discriminative features.[37]

DA-SMOTE: To move the synthetic samples away from the danger area, the DA-SMOTE designs an attention module to automatically learn an adaptive interpolation factor from the data distribution. In the process of learning, the interpolation factor can make the synthetic sample dynamically stay off the center of normal samples, thereby further reducing the risk of falling into the danger area.

Fig.4(b) shows the architecture of the attention module, as well as the generation procedure of the DA-SMOTE. Inspired by the research[38], the attention module is sequentially composed of a dimension-reduction linear layer (with reduction ratio r), a ReLU, a dimension-increasing linear layer, a ReLU, and a linear layer together with a Sigmoid activation at the end. Then, the generation process is briefly described as follows. First, for each abnormal sample z, it is concatenated with one of its k nearest neighbors $\tilde{z}$ to form a sample pair$[z, \tilde{z}]$. Second, $[z, \tilde{z}]$ is fed into the attention module, and its scalar output θ before the Sigmoid activation is expressed by

$$\theta = W_3\left(\varphi\left(W_2\varphi(W_1[z, \tilde{z}])\right)\right) \tag{5}$$

where W_1, W_2, W_3 represent the parameter matrices of three linear layers, and φ refers to the ReLU function. It is worth noting that the attention module requires enough learning ability so as to learn a proper interpolation factor for each sample pair[38]. Third, θ is transformed to the range of $(0, 1)$ using the Sigmoid activation[5], which can be expressed by

$$\hat{\alpha} = 1/\left(1 + e^{-\theta}\right) \tag{6}$$

where $\hat{\alpha}$ is the learnable interpolation factor. Fourth, the synthetic sample can be expressed by

$$z_{new} = z + \hat{\alpha}(\tilde{z} - z) \tag{7}$$

Finally, above steps are repeated until the data set is balanced.

Decoder Network: The Decoder network aims to map the latent features back to the original space. As depicted in Fig.4(a), the Decoder network is also composed of multiple Transformer Encoder layers. Furthermore, to ensure that the dimensions of the output are consistent with those of the input, the Decoder network adds a linear layer at the end.

3.3.3. Training Objective of the Developed DA-SMOTE-ED

The training objective of the developed DA-SMOTE-ED needs to consider the following aspects: on one hand, the learnable features not only keep the main information of the input but also reduce the inter-class overlap; on the other hand, the learnable interpolation factors make the synthetic samples stay off normal samples. Therefore, this paper considers two losses (i.e., reconstruction loss and triplet-center loss), and each loss has its own contribution. And they are respectively introduced as follows.

Reconstruction Loss: To keep the main information of the input (i.e., the data after under-sampling and before data augmentation), this paper adopts the class-balanced l_2-norm as the reconstruction loss $\mathcal{L}_R$, which is expressed by

$$\mathcal{L}_R = \frac{1}{|X|}\sum_{x_i \in X} \frac{1-\beta}{1-\beta^{n_i}} \|x_i - \hat{x}_i\|_2 \tag{8}$$

where x_i is the i-th sample in the set X, and $\hat{x}_i$ is the reconstructed output of x_i. $|X|$ is the number of samples in X, and $\|\cdot\|_2$ is l_2-norm. According to the research[39], the hyper-parameter β is set to 0.999. Moreover, if x_i is a normal sample, $n_i = K$; otherwise, $n_i = A$.

Triplet-Center Loss: In addition to the reconstruction loss, this paper also utilizes the triplet-center loss to guide the learning of latent features and interpolation factors, which is expressed by

$$\mathcal{L}_{TC} = \frac{1}{|Z \cup Z_{new}|}\sum_{z_i \in Z \cup Z_{new}} max\left(\|z_i - c_i\|_2 + d - \left\|z_i - c_{p_i}\right\|_2, 0\right) \tag{9}$$

where z_i is the i-th element in the set $Z \cup Z_{new}$. According to the research[40], the hyper-parameter d is set to 5. Here, if z_i is normal, c_i is the center of normal samples, and c_{p_i} is the center of abnormal samples; if z_i is abnormal, c_i is the center of abnormal samples, and c_{p_i} is the center of normal samples.

Fig.5(a)-(c) shows the changing of sample distribution during the process of training. Obviously, the changes of the normal, abnormal, and synthetic samples are different, and then are separately introduced. First, as shown in Fig.5(d), if z_i is a normal sample, it gradually moves to the center of normal samples while stays off that of abnormal samples. Second, as shown in Fig.5(e), if z_j is an abnormal sample, it gradually stays off the center of normal samples while moves to that of abnormal samples. Third, as shown in Fig.5(f), if z_{new} is a synthetic sample, it gradually stays off the center of normal samples while moves to that of abnormal samples along the line. Moreover, Fig.5(c) indicates that samples from the same class are close to each other, while samples from the different classes are far away from each other (i.e., the intra-class compactness and inter-class separability are enhanced in the learned feature space).

Therefore, this paper uses the reconstruction loss and the triplet-center loss simultaneously to train the DA-SMOTE-ED. The total loss can be expressed by

$$\mathcal{L}_{Total} = \mathcal{L}_R + \mathcal{L}_{TC} \tag{10}$$

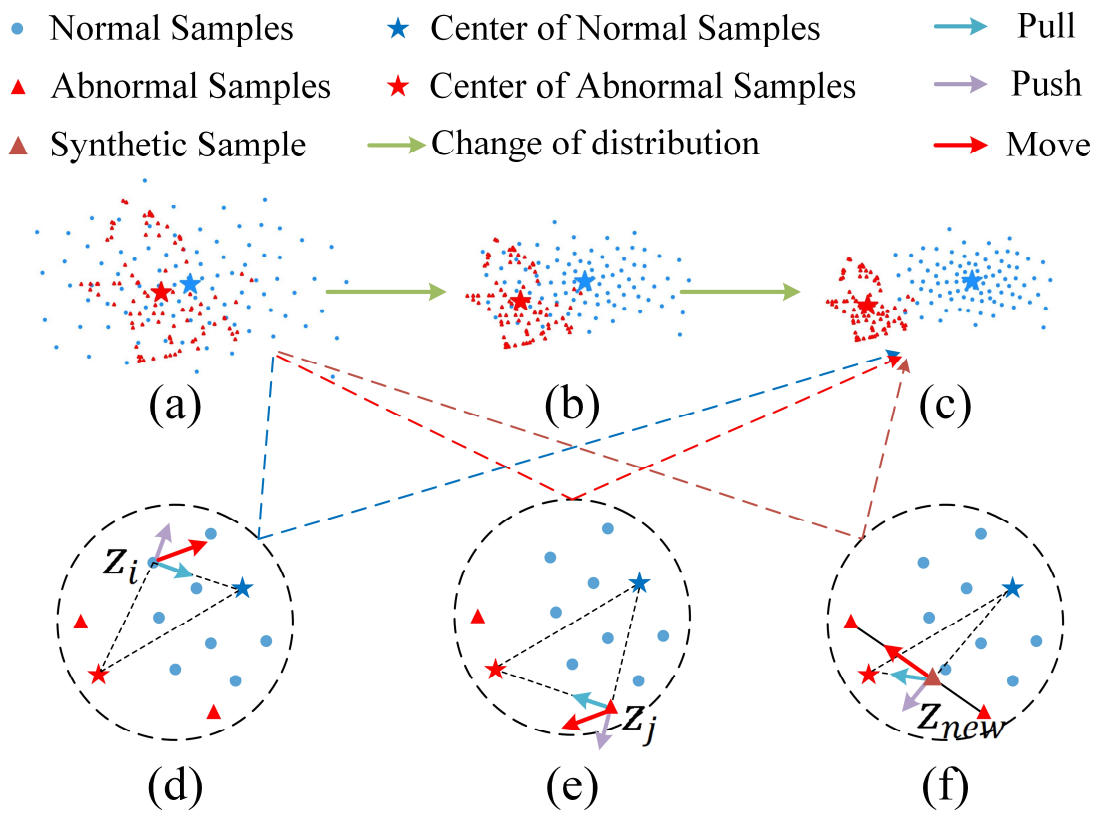

Fig. 5 2D sketch of samples (a) before training the DA-SMOTE-ED, (b) during training the DA-SMOTE-ED, and (c) after training the DA-SMOTE-ED; (d) pulling normal samples to the center of normal samples and pushing them away from the center of abnormal samples, (e) pulling abnormal samples to the center of abnormal samples and pushing them away from the center of normal samples, (f) pulling synthetic samples to the center of abnormal samples and pushing them away from the center of normal samples in the learned feature space.

4. Experimental Results

4.1. Experimental setup

4.1.1. Dataset collection

In this paper, to evaluate the proposed framework, real-world monitoring data of gas turbines was collected from an Asia airline during 2008-2015. For the convenience of understanding, Fig.6(a) shows that the structure and sensor arrangement of the research object, which is mainly composed of the fan, low-pressure compressor (LPC), high-pressure compressor (HPC), combustion chamber (CC), high-pressure turbine (HPT), and low-pressure turbine (LPT).

Research has shown that some gas-path performance parameters can reflect the health status of gas turbines[41], mainly including exhaust gas temperature (EGT), low-pressure rotor speed (N1), high-pressure rotor speed (N2), and fuel flow (FF). However, they are affected by flight conditions, operating environments, etc. To alleviate the issue, the original equipment manufacturer (OEM) transforms them into the exhaust gas temperature margin (EGTM), delta exhaust gas temperature (DEGT), delta core speed (DN2), and delta fuel flow (DFF), and then utilizes the transformed values to infer health status of gas turbines.

Furthermore, Fig.6(b) shows the sample construction process using a sliding window method (window size $l = 10$, step length $s = 10$)[41]. As summarized in Table I, 954 normal samples and 63 abnormal samples are obtained at the end. Then, this paper adopts five-fold cross-validation to divide the gas turbine data set.

TABLE I. Summary of the gas turbine dataset

Five-fold cross-validation	Number of training samples		Number of testing samples	
	Normal	Abnormal	Normal	Abnormal
1st-Fold	763	50	191	13
2nd-Fold	763	50	191	13
3rd-Fold	763	50	191	13
4th-Fold	763	51	191	12
5th-Fold	764	51	190	12

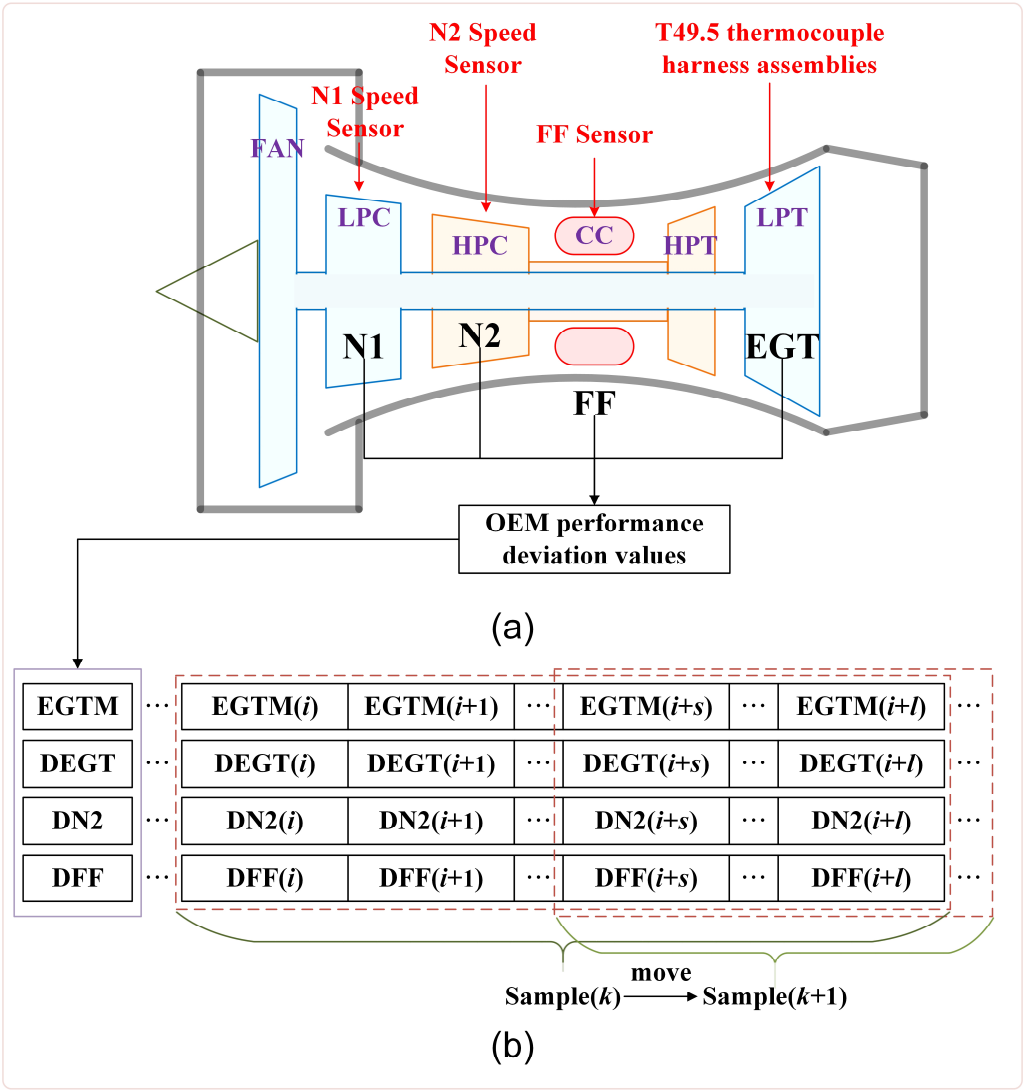

Fig 6 (a) Acquisition and preprocessing flow of gas turbine performance data, (b) sample construction using a sliding window method, where Sample(k) denotes k-th sample.

4.1.2. Implementation details

Experiments are implemented in Python 3.9 with PyTorch 1.11.0 using an Intel Core i5-12400F, 16G DDR4 processor. The hyper-parameters are set according to the recommendations [21],[38],[41]. First, Adam adopts a learning rate lr =0.0002, momentums $(\beta_1, \beta_2) = (0.5, 0.999)$, and batch size=128. Second, the number of clusters in the clustering-based under-sampling is 90. Third, the Encoder network consists of 2 Transformer Encoder layers, in which $h = 4$ and $d_i = 128/4 = 32$[41]. The DA-SMOTE adopts the number of nearest neighbors $k = 5$[21] and a reduction ratio $r = 16$[38]. The Decoder network consists of 2 Transformer Encoder layers ($h = 4$, $d_i = 32$) and 1 linear layer[41]. Finally, the classifier also consists of 2 Transformer Encoder layers ($h = 4$, $d_i = 32$) and 1 linear layer with a Softmax.

4.1.3. Evaluation metrics

In this paper, true positive rate (TPR), true negative rate (TNR), and balanced accuracy are used to evaluate the detection performance of the methods on abnormal samples, normal samples, and overall samples, respectively[42]. And their formula can be expressed by

$$TPR = N_{11}/(N_{11} + N_{12}) \tag{11}$$

$$TNR = N_{22}/(N_{21} + N_{22}) \tag{12}$$

$$balanced\ accuracy = (TPR + TNR)/2 \tag{13}$$

where N_{11}, N_{12}, N_{21}, and N_{22} is summarized in Table II.

TABLE II. An illustration of confusion matrix

Actual Class	Predicted Class	
	Abnormal	Normal
Abnormal	N_{11}	N_{12}
Normal	N_{21}	N_{22}

4.2. Validating the effectiveness of the developed method

This paper selects 1 baseline method without any re-sampling method and 8 popular re-sampling methods for comparison, which are summarized as follows.

(1) **Baseline method:** The SANet is directly trained on the imbalanced training set, and is used to classify the test set.

(2) **8 re-sampling methods:** These methods are divided into classical re-sampling methods and deep learning-based re-sampling methods. Classical re-sampling methods include four over-sampling methods: ADASYN[43], SMOTE[15], Borderline- SMOTE (BSMOTE)[24], and KMeans-SMOTE (KSMOTE)[25], and two popular hybrid re-sampling methods: SMOTE-Tomek[26] and SMOTE-ENN[26]. Deep learning-based re-sampling methods include variational auto-encoder (VAE)[44] and generative adversarial network (GAN)[45]. More specifically, the balanced training set is obtained using the re-sampling method. Then, the SANet is trained on the balanced training set, and is used to classify the test set.

Figs.7-8 show the TPR and TNR of the considered methods. First, for the TPR, it can be observed that the developed method outperforms the baseline method and 8 re-sampling methods in all of the tests. Significantly, its TPR is close to 100% in the 2nd-fold experiment, which is 13.08% higher than the SMOTE-ENN. Second, it can be seen that the TNR of the developed method exceeds 91.83%, yet is inferior to the comparison methods. The reason is that the comparison methods pay more attention to normal samples and ignore abnormal samples, which is obviously not suitable for gas turbines with extremely strict safety requirements.

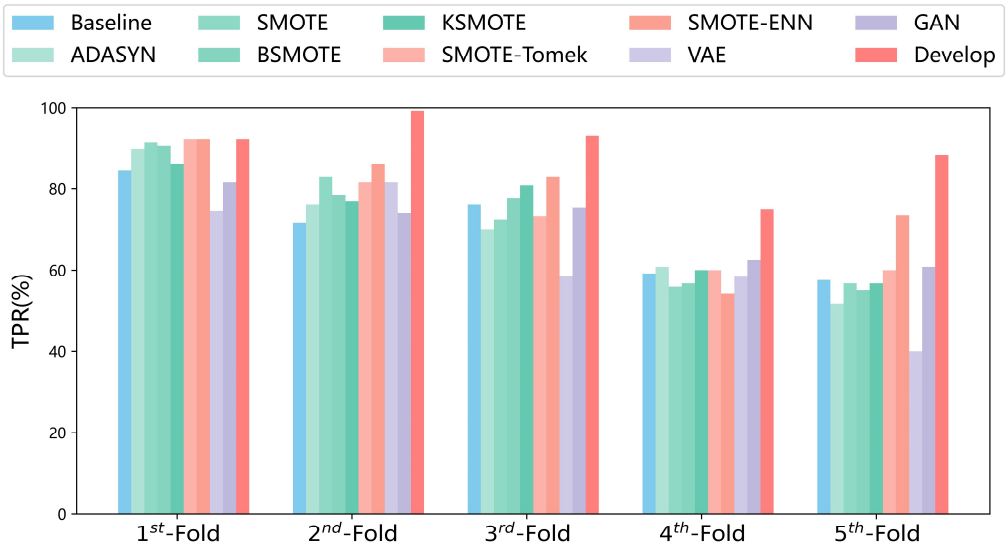

Fig 7 TPR of different anomaly detection models.

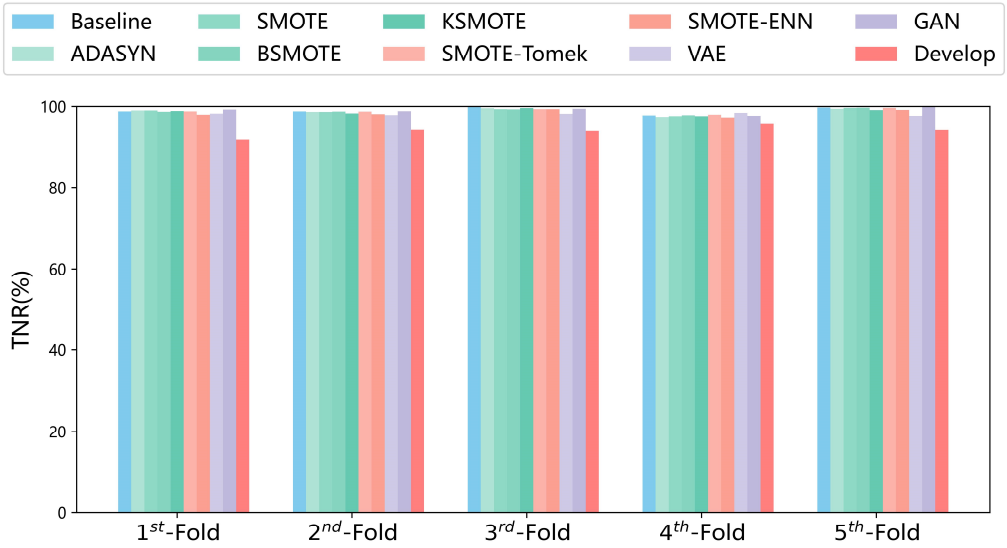

Fig 8 TNR of different anomaly detection models.

Further, the averages of TPR, TNR, and balanced accuracy are summarized in Table III, which can be concluded as follows.

(1) In terms of average TPR and balanced accuracy, 7 re-sampling methods outperform the baseline method. And, for average TNR, all 8 re-sampling methods are very close to the baseline method. Obviously, a proper re-

sampling method can improve the classification performance of the classifier on the imbalanced data set.

(2) In terms of average TPR and balanced accuracy, the SMOTE-Tomek and SMOTE-ENN are superior to the ADASYN, SMOTE, BSMOTE, KSMOTE, VAE, and GAN. And, for average TNR, the aforementioned methods are very similar. It can be observed that the performance improvements of the SMOTE-Tomek and SMOTE-ENN may be due to the advantage of the hybrid-sampling strategy.

(3) Compared with the considered comparison methods, the developed method achieves the best results in terms of average TPR and balanced accuracy. Especially compared with SMOTE-Tomek and SMOTE-ENN, the average balanced accuracy is improved by 5.63% and 3.67%, respectively, which indicates that the performance of our developed hybrid re-sampling method is superior.

TABLE III. Average TPR, TNR, balanced accuracy, and inference time of different anomaly detection methods

Method	TPR (%)	TNR (%)	Balanced Accuracy (%)	Inference Time (milliseconds)
Baseline	69.79	98.98	84.39	**1.399**
ADASYN	69.73	98.82	84.27	1.505
SMOTE	71.88	98.85	85.37	1.423
BSMOTE	71.72	98.85	85.28	1.515
KSMOTE	72.10	98.71	85.41	1.422
SMOTE-Tomek	73.38	98.90	86.14	1.419
SMOTE-ENN	77.81	98.39	88.10	1.432
VAE	62.59	98.11	80.35	1.408
GAN	70.82	**99.02**	84.92	1.536
Developed	**89.59**	93.95	**91.77**	1.473

In the test stage, all considered methods only utilize the trained SANet with the same hyper-parameters to infer test samples. Thus, the inference time is theoretically the same. Table III shows the inference time of all considered methods. Obviously, each inference time is approximately equal, and slight differences may be caused by random factors. Moreover, the developed method takes only 1.473 milliseconds to infer 204 test samples, which is acceptable in engineering.

4.3. Investigation on performance with different numbers of clusters

Tables IV-VI show the average of TPR, TNR, and balanced accuracy of the developed method using different numbers of clusters K, respectively. It can be seen that as the number of clusters decreases, the detection performance of the developed method gradually becomes weaker for normal samples, while becomes stronger for abnormal samples. However, the proposed method still retains high comprehensive detection performance. When the number of clusters is 90, the developed method achieves the best average balanced accuracy. Moreover, when $K = 50$ (i.e., the number of normal samples is the same as that of abnormal samples), the developed method becomes a clustering-based under-sampling method. The average balanced accuracy becomes worse when $K = 50$, which also validates the effectiveness of the developed DA-SMOTE-ED.

TABLE IV. TPR (%) of the developed method using different numbers of clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	92.31	99.23	84.62	71.67	91.67	87.9
60	90.77	95.38	94.62	73.33	91.67	89.15
70	93.08	100	89.23	75	94.17	**90.29**
80	90	99.23	84.62	75	90.83	87.94
90	92.31	99.23	93.08	75	88.33	89.59
100	89.23	97.69	90.77	73.33	86.67	87.54
110	90.77	97.69	87.69	73.33	82.5	86.4
120	92.31	96.92	85.38	75	74.17	84.76
130	91.54	93.08	86.92	75.83	79.17	85.31
140	91.54	99.23	82.31	78.33	72.5	84.78
150	89.23	96.92	90	83.33	65.83	85.06

TABLE V. TPR (%) of the developed method using different numbers of clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	90.73	90.26	87.75	94.87	88.37	90.4
60	93.56	92.2	90	95.18	92.63	92.71
70	93.51	91.73	92.2	95.08	90.84	92.67
80	93.3	93.09	93.61	95.86	93.79	93.93
90	91.83	94.19	93.93	95.65	94.16	93.95
100	93.51	94.03	95.29	95.6	95.37	94.76
110	95.03	95.13	96.13	95.81	95.37	95.49
120	94.66	95.08	96.13	95.76	94.21	95.17
130	95.97	95.86	96.75	95.71	96.95	96.25
140	96.6	95.92	96.44	96.39	97.05	**96.48**
150	95.76	96.13	95.29	96.54	97.47	96.24

TABLE VI. Balanced accuracy of the developed method using different numbers of clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	91.52	94.75	86.18	83.27	90.02	89.15
60	92.16	93.79	92.31	84.26	92.15	90.93
70	93.29	95.86	90.71	85.04	92.5	91.48
80	91.65	96.16	89.11	85.43	92.31	90.93
90	92.07	96.71	93.5	85.33	91.25	**91.77**
100	91.37	95.86	93.03	84.47	91.02	91.15
110	92.9	96.41	91.91	84.57	88.93	90.94
120	93.48	96	90.76	85.38	84.19	89.96
130	93.75	94.47	91.84	85.77	88.06	90.78
140	94.07	97.57	89.37	87.36	84.78	90.63
150	92.5	96.52	92.64	89.94	81.65	90.65

4.4. Investigation on the intra-class compactness and inter-class separability

This paper uses t-distributed stochastic neighbor embedding technology to provide 2-dimensional visualizations of training samples and re-sampled training samples. First, Fig.9(a) indicates that training samples are highly scattered and overlapped in the original space. Second, Fig.9(b) shows that the cluster-based under-sampling method reduces the degree of class imbalance. Third, Fig.9(c) illustrates the distribution of under-sampled training samples before data augmentation in the learned feature space, in which samples of the same class are clustered with one another, while samples of different classes are far away from one another. Therefore, the Encoder network can enhance the intra-class compactness and inter-class separability. Finally, the distribution of training samples after under-sampling and data augmentation in the learned feature space is depicted in Fig.9(d). It can be observed that the DA-SMOTE-ED generates synthetic samples that stay off the danger area.

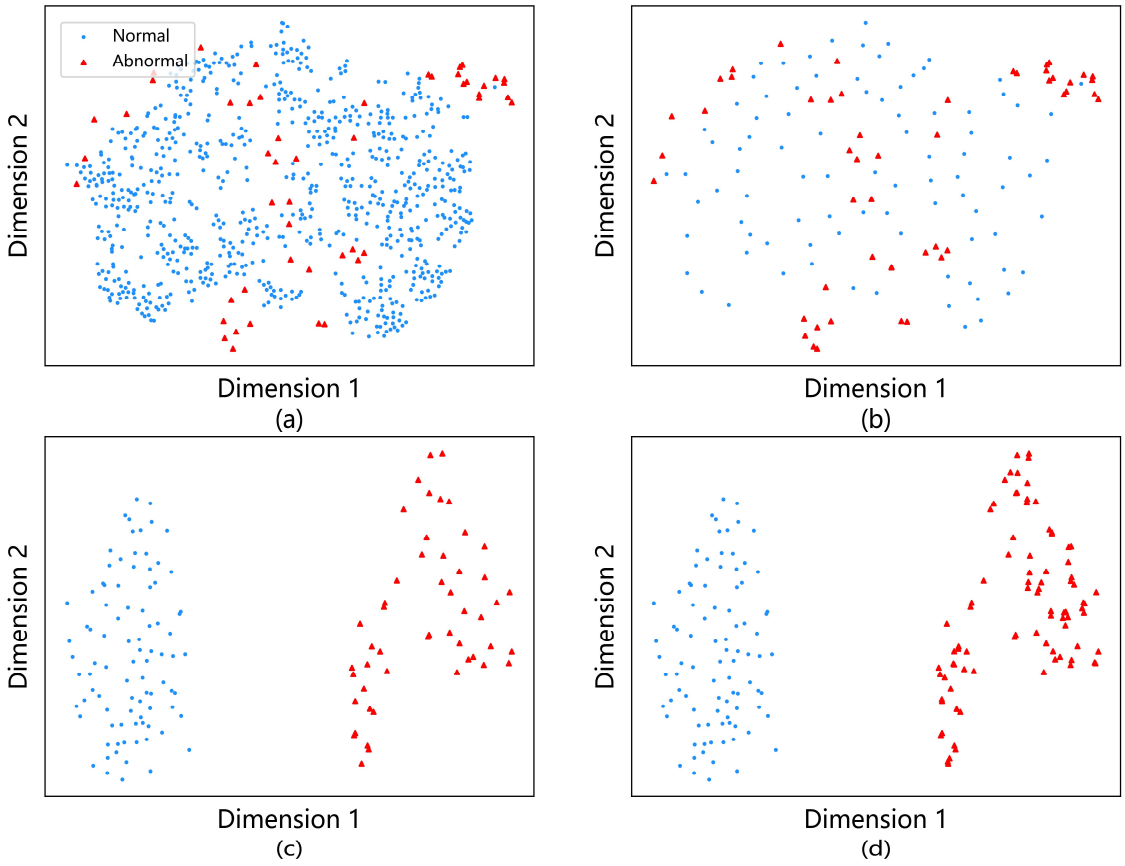

Fig. 9 2D sketch maps of (a) training samples in the original data space; (b) training samples after clustering-based under-sampling in the original data space; (c) training samples after clustering-based under-sampling in the learned feature space; (d) training samples after both clustering-based under-sampling and DA-SMOTE-ED-based data augmentation in the learned feature space.

5. Conclusion

To balance the imbalanced data under the condition of inter-class overlap, this paper develops a novel attention-based data augmentation method, namely DA-SMOTE-ED, which serves as a key step in anomaly detection. First, the Encoder network maps the under-sampled training set to a class-separable feature space. Second, the DA-SMOTE assigns proper interpolation factors to abnormal samples and their nearest neighbors to generate high-quality synthetic

samples. Third, the Decoder network maps synthetic samples back to the original space, and augmented samples are put together with under-sampled samples as the balanced data set.

The effectiveness of our re-sampling method has been verified using a real-world gas turbine data set. In the experiment, compared with the traditional hybrid re-sampling methods SMOTE-Tomek and SMOTE-ENN, our re-sampling method achieves improvements of 5.63% and 3.67% in terms of the average balanced accuracy. The reasons are that, on one hand, the clustering-based under-sampling method retains important and useful information; on the other hand, the DA-SMOTE-ED balances the under-sampled training set and reduces the risk of generating noise data.

Finally, anomaly detection of gas turbines is an evolving research topic. Our future work will focus on developing new hybrid re-sampling methods to further improve the balanced accuracy of anomaly detection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: